

Solar C appendix

Sun · C · downloadable appendix

Solar carbon: 8.460 ± 0.068

Asplund, Amarsi & Grevesse 2021: 8.46 ± 0.04 · this measurement +0.000 dex

HARPS · VIS · Reference Grade · SET-AGSS21 · ENGINE-A-3DNLTE · $n = 3$. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

8.460 ± 0.068

± 0.018 stat · ± 0.041 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar_harps_molecfits_corrected · Reference Grade

red-optical

8.536 ± 0.077

± 0.045 stat · ± 0.050 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar_kpno_kurucz2005_corrected · Reference Grade

Error-bar Forest Plot

VIS — harps_solar_harps_molecfits_corrected A(C) dex; axis 8.17-8.79

ATOM-CI_5052 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.369 ± 0.092 stat ± 0.088 syst
ATOM-CI_5380 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.526 ± 0.097 stat ± 0.234 syst
MOL-C2_Swan · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.390 ± 0.009 stat ± 0.120 syst
MOL-CH_Gband · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=1		8.482 ± 0.005 stat ± 0.076 syst
SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=3		8.480 ± 0.018 stat ± 0.041 syst
SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied Reference Grade · ALL · n=3		8.460 ± 0.018 stat ± 0.041 syst
SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=3		8.473 ± 0.019 stat ± 0.041 syst
indicator not supplied · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied Codex Grade · ALL · n=2		8.476 ± 0.030 stat ± 0.145 syst
indicator not supplied · C I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Codex Grade · ALL · n=2		8.488 ± 0.032 stat ± 0.145 syst

VIS — iag_fts_solar_atlas solar_iag A(C) dex; axis 8.17-8.79

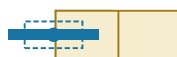
indicator not supplied · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Codex Grade · ALL · n=2		8.372 ± 0.116 stat ± 0.041 syst
indicator not supplied · C I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied Codex Grade · ALL · n=2		8.353 ± 0.116 stat ± 0.041 syst
indicator not supplied · C I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied Codex Grade · ALL · n=2		8.365 ± 0.115 stat ± 0.041 syst

VIS — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(C) dex; axis 8.17-8.79

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=3		8.437 ± 0.065 stat ± 0.041 syst
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VIS — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(C) dex; axis 8.17-8.79

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=3



8.417
±0.065 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=3



8.430
±0.064 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=2



8.345
±0.105 stat ±0.145 syst

indicator not supplied · C I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2



8.357
±0.103 stat ±0.145 syst

VIS — kpno_solar_atlas solar_kpno_molecfit_corrected A(C) dex; axis 8.17-8.79

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=3



8.450
±0.067 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=3



8.430
±0.068 stat ±0.041 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=3



8.443
±0.067 stat ±0.041 syst

indicator not supplied · C I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=2



8.353
±0.108 stat ±0.145 syst

indicator not supplied · C I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2



8.365
±0.107 stat ±0.145 syst

VIS axis: 8.17 · 8.32 · 8.48 · 8.63 · 8.79 A(C) dex

red-optical — iag_fts_solar_atlas solar_iag A(C) dex; axis 8.39-8.69

SET-AGSS21 · C I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=10



8.526
±0.045 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=10



8.485
±0.042 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=10



8.488
±0.041 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



8.566
±0.038 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=6



8.528
±0.040 stat ±0.050 syst

red-optical — iag_fts_solar_atlas solar_iag A(C) dex; axis 8.39-8.69

indicator not supplied · C I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=6



8.527
±0.039 stat ±0.050 syst

red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(C) dex; axis 8.39-8.69

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH ·
ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=10



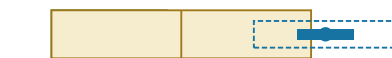
8.536
±0.045 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=10



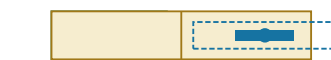
8.532
±0.044 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



8.610
±0.020 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=6



8.568
±0.020 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=6



8.563
±0.019 stat ±0.050 syst

red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(C) dex; axis 8.39-8.69

SET-AGSS21 · C I · Synth · 3D-NLTE · Amarsi · SYNTH ·
ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=10



8.529
±0.058 stat ±0.050 syst

SET-AGSS21 · C I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=10



8.531
±0.056 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=6



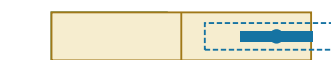
8.610
±0.023 stat ±0.050 syst

indicator not supplied · C I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=6



8.575
±0.025 stat ±0.050 syst

indicator not supplied · C I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=6



8.576
±0.025 stat ±0.050 syst

red-optical axis: 8.39 · 8.46 · 8.54 · 8.61 · 8.69 A(C) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

How this was measured. One row per band. The indicator set is what was fitted; the holding is the solar spectrum it was fitted against; the engine and treatment say how. Dominant term is the largest single contributor to the reported uncertainty, and gf rung is the pedigree of the oscillator strengths that pool could reach.

Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, C₂ and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

[C/O indicator strategy](#) · [Methodology](#)

Download the evidence

[Download PDF — Solar C appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

a8369474dbef157366320260731ced27933bec44

Feed C.json v1.76 · updated 2026-10-02T03:08:12Z

Generated 2026-10-01T21:08:32-06:00 (source commit timestamp)

Feed SHA-256 ddddacb7675a5de67417c2cfa74bdf8b3e430f26e62a7bc5cf322d20d5121648

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Atomic data and gf authorities. The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [\[3\]](#), [\[4\]](#).

Atomic data and gf authorities. AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign's NIST/MCHF adjudication; no gf source is substituted here. Sources: [\[5\]](#).

Model and grid sources. Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [\[6\]](#).

Atomic data and gf authorities. Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [\[3\]](#), [\[4\]](#).

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

Molecular line data and delivery. C2 line-data delivery: C2 Swan $d^3\Pi_g-a^3\Pi_u$ (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [8].

Molecular line data and delivery. CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest's MNRAS locator conflicts with that record. Sources: [2], [7], [9].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 Å) Sources: [6].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 Å) Sources: [6].

Product-to-source index

C-products-45eae148b981 · products · VIS · solar_harps_molecfit_corrected · ATOM-Cl_5052 · 1D-LTE · not supplied · [1], [3], [4], [7]

C-products-7718d3380b98 · products · VIS · solar_harps_molecfit_corrected · ATOM-Cl_5380 · 1D-LTE · not supplied · [1], [3], [4], [7]

C-products-8614c00a6f61 · products · VIS · solar_harps_molecfit_corrected · MOL-C2_Swan · 1D-LTE · not supplied · [1], [2], [7], [8]

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C-products-475299546416 · products · VIS · solar_harps_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [7]

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C-products-5c2d1868fca6 · products · VIS · solar_iag · not supplied · ENGINE-A · not supplied · [1], [3], [4], [6]

C-products-750df87348c4 · products · VIS · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [7]

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C-products-fa2a9a61f2af · products · VIS · solar_kpno_molecfite_corrected · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)

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C-products-246b5509c29e · products · red-optical · solar_iag · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)

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Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Li, Amarsi, Papoulia, Ekman & Jonsson 2021, *MNRAS* 502, 3780 DOI: 10.1093/mnras/stab214. [source record](#)

Model and grid sources

[6] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. *A&A* 630, A104 DOI: 10.1051/0004-6361/201936265. [source record](#)

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